

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

.Net Core and Progressive Web Application Practical

Practical – VI

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Class: TYIT	Batch: 1
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A) Create a simple website for a College Information Portal and convert it into a Progressive Web Application.

Code:-

service-worker.js:-

```
const cache = "v1";

self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(cache).then(cache => {
      return cache.addAll([
        "/",
        "/index.html"
      ]);
    })
  );
});
console.log("Service Worker Installed");
});

self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() => {
      return caches.match("/index.html");
    })
  );
});
});
```

Manifest.json:-

```
{
"name": "College Information Portal",
"short_name": "College Portal",
"start_url": "/",
"display": "standalone",
```

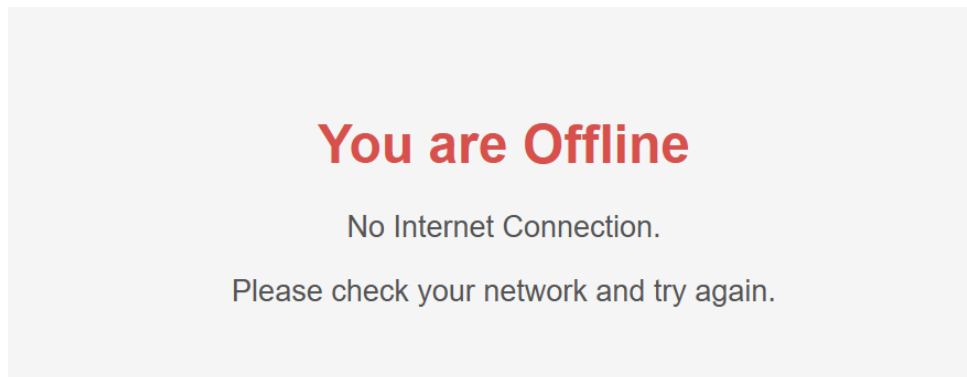
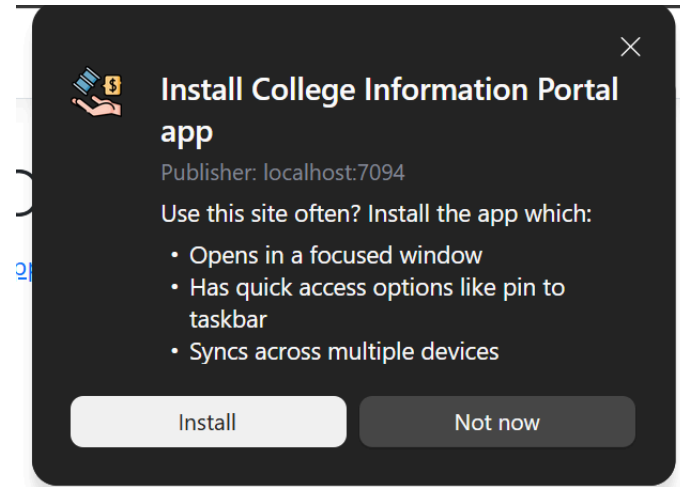
```
"icons": [  
  {  
    "src": "icons/icon1.png",  
    "sizes": "128x128",  
    "type": "image/png"  
  },  
  {  
    "src": "icons/icon2.png",  
    "sizes": "512x512",  
    "type": "image/png"  
  }  
]  
}
```

Index.html:-

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="utf-8" />  
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />  
  <title>Offline</title>  
  <style>  
    body {  
      font-family: Arial, sans-serif;  
      text-align: center;  
      margin-top: 100px;  
      background-color: #f5f5f5;  
    }  
  
    h1 {  
      color: #d9534f;  
    }  
  
    p {  
      color: #555;  
      font-size: 18px;  
    }  
  </style>  
</head>  
<body>  
  <h1>You are Offline</h1>  
  <p>No Internet Connection.</p>  
  <p>Please check your network and try again.</p>
```

```
</body>
</html>
```

Output:-



B) Create a PWA that displays a custom offline page when the user is not connected to the internet.

Code:-

Manifest.json:-

```
{
  "name": "Html to Pwa",
  "short_name": "Htp",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
```

```

    "src": "icons/icon2.png",
    "sizes": "512x512",
    "type": "image/png"
  }
]
}

```

service-worker.js:-

```

self.addEventListener('install', e => {
  console.log("Service Worker Installed");
});

```

```

self.addEventListener('fetch', e => {
  e.respondWith(fetch(e.request));
});

```

Layout.cshtml:-

```

<link rel="manifest" href="/manifest.json"/>
</head>
<script>
  if('serviceWorker' in navigator) {
    navigator.serviceWorker.register('/service-worker.js');
  }
</script>

```

Output:-

